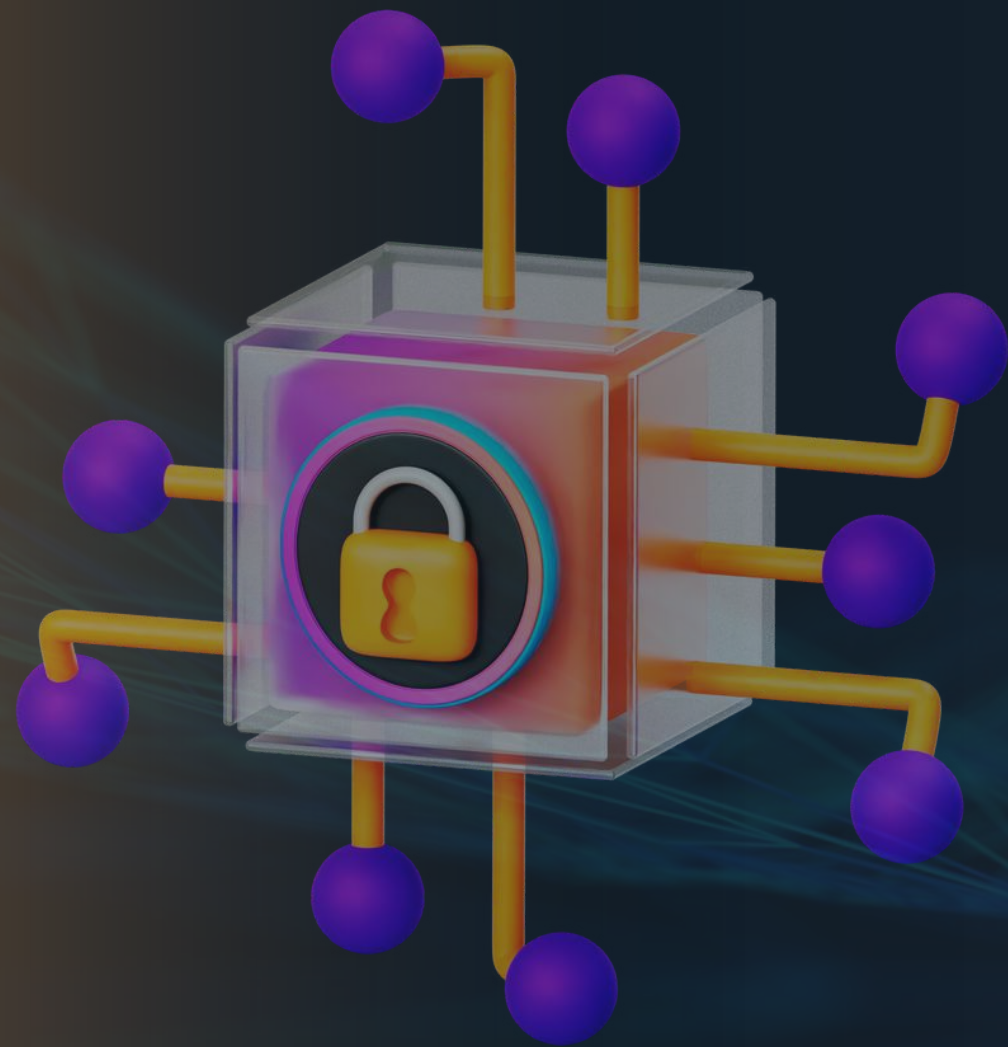




University of Sri Jayewardenepura
Faculty of Technology

A BLOCKCHAIN-BASED UNIFIED DIGITAL IDENTITY SYSTEM WITH BIOMETRIC SECURITY – IN THE SRI LANKAN CONTEXT

Project Reminder



Unified Digital Identity for Sri Lanka

- Each citizen has a single digital identity usable across all sectors
- Eliminates separate IDs like NIC, university ID, hospital ID
- Provides consistent and reliable identity verification

Blockchain+ Biometric Authentication

- Blockchain stores identity records securely and immutably
- Biometric verification confirms the real owner of the identity
- Removes dependency on passwords or manual checking

Replace Multiple IDs with One Secure ID

- One trusted identity for government, banking, healthcare, and education
- Reduces fraud and impersonation
- Faster and more convenient service access

Response to Examiners' Comments

Why Blockchain

- Evaluated centralized, federated, and distributed alternatives
- Blockchain chosen for decentralization, immutability, traceability, and secure auditing

Novelty Added

- Biometric Digital Twins for adaptive offline proofs
- Fully cardless identity verification
- Multi modal biometrics (fingerprint + face)
- Decentralized identity storage

Cost Reduction Actions

- Use open source blockchain platforms
- Cloud deployment instead of local infrastructure
- Pilot implementation before nationwide rollout
- Integration with existing government systems
- Modular scalable architecture



Research Objectives



Main Objective

To design and evaluate a blockchain-based digital identity framework integrated with biometric verification for secure and unified identity management.

Specific Objectives



Provide single identity



Biometric authentication



Decentralized verification



Prevent fraud



Ensure interoperability

Objective Coverage Status

Objective

Single digital identity

Biometric authentication

Blockchain storage

Decentralization

Interoperability

Status



Evidence

User registration module

Fingerprint/face login demo

Smart contract + ledger

Local private network

API integration next phase

System Overview



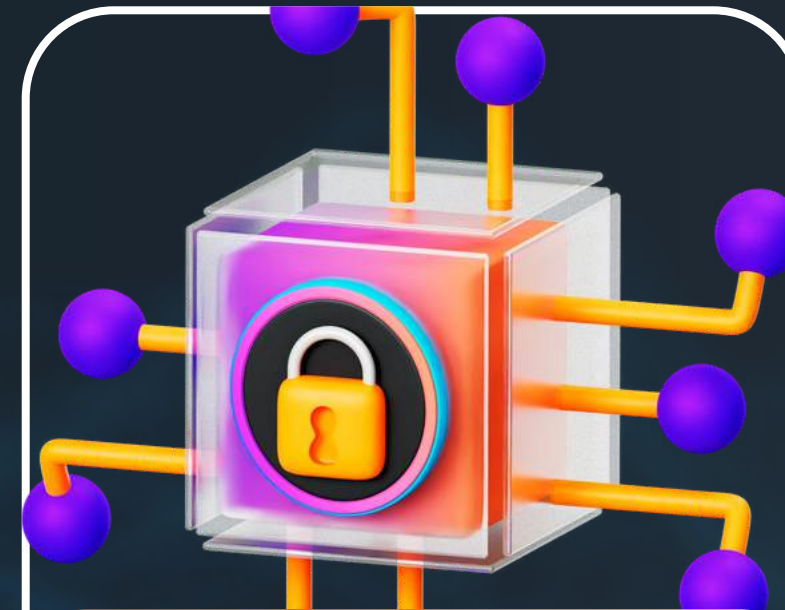
User

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Biometric

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Blockchain

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Service access

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System Architecture



Technologies Implemented

-  Blockchain network
-  Blockchain network
-  Blockchain network
-  Blockchain network



Challenges Faced & Solutions

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➞ Difficult to use

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➞ Not well known yet

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➔ More people will use it

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➔ Technology will get better

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Future Work

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Thank You

